

Linear Algebra (MTS203)

Midterm Exam, (120 mins — 20 Marks)

Spring 2024 Class no. 40241

Exam Date - March 11, 2024



Question - 1 (3 marks)

Find all solutions to the linear system:

$$x_1 - x_2 + 3x_3 + 2x_4 = 1$$

$$-x_1 + x_2 - 2x_3 + x_4 = -2$$

$$2x_1 - 2x_2 + 7x_3 + 7x_4 = 1$$

Question - 2 (2 marks)

Given:

$$A = \begin{bmatrix} \mathcal{I} & \mathcal{O} & \mathcal{C} \\ \mathcal{B} & \mathcal{I} & \mathcal{O} \\ \mathcal{O} & \mathcal{O} & \mathcal{I} \end{bmatrix}$$

where all of the submatrices are $n \times n$ blocks, determine the block form of A^{-1} .

Question - 3 (3 marks)

Let B and C be $n \times n$ matrices with the property that $B\mathbf{x} = C\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Prove that $B = C$.

Question - 4 (3 marks)

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}$$

where all four blocks are $n \times n$ matrices. If A_{11} , and A_{22} are nonsingular, show that A must also be nonsingular, and that A^{-1} must be of the form

$$\left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline O & A_{22}^{-1} \end{array} \right]$$

Determine C .

Question - 5 (3 marks)

What is an *involution*? Give its formula. Using this formula form 2 different matrices that are their own inverses.

Question - 6 (1 mark)

Suppose that a 3×3 matrix A is factored into a product:

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Determine $|A|$.

Question - 6 (2 mark)

Show that if $\det(A) = 1$ then

$$\text{adj}(\text{adj } A) = A$$

Question - 7 (2 mark)

Let R denote the set of real numbers. Define scalar multiplication by:

$$\alpha x = \alpha \cdot x \text{ (the usual multiplication of real numbers)}$$

and define vector addition $+$, by

$$x + y = \max(x, y) \text{ (the maximum of two numbers } x \text{ and } y)$$

Is R a vector space with these operations? Prove your answer.

Question - 8 (1 mark)

Is the set $\{1, x^2, x^2 - 2\}$ a spanning set for P_3 ? Prove your answer.